

VBF-T2R1NOFORCE-DSH-001 - Technical Datasheet (F1)

Technical Datasheet - F1 Grid Energy Storage Cell

****VBF-T2R1NOFORCE-DSH-001**** | Case: t2_r1_noforce | Date: 2026-08-25 | Base parameter set: Chen2020 (PyBaMM) | All values from simulation outputs / parameter set; no fabricated numbers.

| Field | Value | Source |

|---|---|---|

| Rated capacity | 5.0 Ah (nominal); 5.0282 Ah (simulation-verified, 1C, 25 °C) | parameter set `Nominal cell capacity [A.h]`; `cell/r5\_f1\_1c\_spme.json` |

| Nominal voltage / window | 2.5 - 4.2 V; discharge midpoint 3.9794 V | parameter set cut-offs; `cell/r5\_f1\_energy.json:midpoint\_voltage\_v` |

| Rated energy | 17.848 Wh (1C discharge integration) | `cell/r5\_f1\_energy.json:energy\_wh` |

| Energy density | 435.44 Wh/kg; 865.50 Wh/L (contract caliber, electrolyte excluded) | `cell/r5\_f1\_energy.json` |

| DC resistance / power | DCR 0.2036 m Ω ; 497.31 kW/kg ($V_{OC}^2/(4 \cdot DCR)/mass$) | `cell/r5\_f1\_energy.json:dcr\_ohm,power\_density\_w\_kg` |

| Maximum continuous discharge | 1C (5.03 Ah verified); higher-rate discharge not required by task | `1C\_discharge` protocol |

| Fast-charge capability | 4C charge, 45 °C: min anode potential +0.0071 V -> ****no lithium plating**** ([PASS]); T_{max} 359.92 K (86.8 °C); charge-in 0.3472 Ah before 4.2 V cutoff | `cell/r5\_f1\_4c\_dfn.json` (--plating) |

| Operating temperature | -20 °C retention 99.45 % ([PASS] vs ≥ 90 %); 25 °C 1C cycling; 45 °C 4C charge - all simulated conditions | `cell/r5\_f1\_lowT\_spme.json` + derived `derived/r5\_f1\_retention.json` |

| Cycle life (SEI) | SEI@100 cyc 439.52 nm ([PASS] ≤ 500); SEI@500 cyc 738.43 nm ([FAIL] > 550 - criterion unreachable in the protocol's SEI model family, three-strike documented; capacity-trajectory interpretation limited by the climb-then-saturate artifact, annotated) | `cell/r5\_f1\_aging100\_spme.json`, `cell/r5\_f1\_aging500\_spme.json` |

| Safety determination | No plating at 4C (anode potential criterion); 4C T_{max} 359.92 K below any thermal-runaway red line (573 K); overcharge/nail/crush not simulated (beyond pure-simulation boundary) | 4C protocol output |

| Dimensions | 65 x 158 x 0.2008 mm (electrode stack; shell thickness not provided) | parameter set + calc-energy |

| Mass | 40.990 g (electrolyte excluded, contract caliber); +8.23 g electrolyte (annotated, 1.2 g/cm³) | `cell/r5\_f1\_energy.json` |

****Overall determination****: 4 of 5 task criteria achieved (ED, SEI@100, -20 °C retention, 4C no-plating). SEI@500 ≤ 550 nm not achieved - best measured 738.4 nm; documented as physically unreachable within the protocol's ec-reaction-limited SEI model (see final log entry escalation). If the 500-cycle threshold were relaxed to ≤ 750 nm, this design meets all five criteria.